Exercise 2
Large Signal & Small Signal Analysis

Winter Term 2025/26

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Exercise 2 – Task 1.1

- 1.1 Calculate the gate-source and gate-drain capacitances C_{GS} and C_{GD} of a transistor in the linear and saturation region (C'_{ox} = 12 fF/µm²). Overlap capacitances are to be neglected!

$$\frac{W}{L} = \frac{10\ \mu\mathrm{m}}{10\ \mu\mathrm{m}}$$

$$C_{\mathrm{ox}} = C'_{\mathrm{ox}}\, WL = 12\,\frac{\mathrm{fF}}{\mu\mathrm{m}^2}\,100\ \mu\mathrm{m}^2 = 1.2\ \mathrm{pF}$$

- linear region(C_{GSov} & C_{GDov} neglected)

  $C_{\mathrm{GS}} \approx 0.5 C_{\mathrm{ox}} \approx 0.6\ \mathrm{pF}$

  $C_{\mathrm{GD}} \approx 0.5 C_{\mathrm{ox}} \approx 0.6\ \mathrm{pF}$

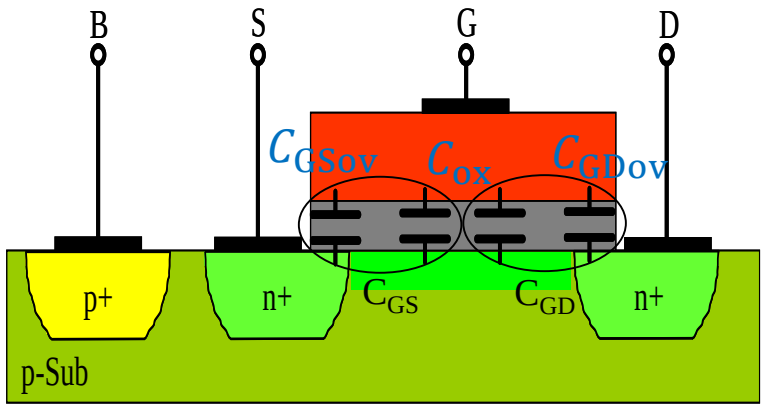
- saturation region (C_{GSov} & C_{GDov} neglected)

  $C_{\mathrm{GS}} \approx 0.6 C_{\mathrm{ox}} \approx 0.72\ \mathrm{pF}$

  $C_{\mathrm{GD}} \approx 0$

Exercise 2 – Task 1.2

- 1.2 Briefly explain why the gate-drain capacitance differ in the linear and the saturation region!



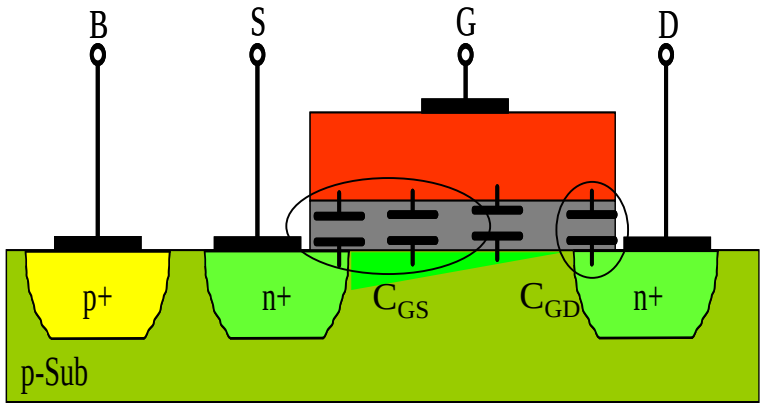
linear region

$C_{GS} \approx 0.5C_{ox} + C_{GSov}$

$C_{GD} \approx 0.5C_{ox} + C_{GDov}$

Symmetric channel shape in the linear region

C_{OX} is distributed evenly between drain and source.



saturation region

$C_{GS} \approx 0.6C_{ox} + C_{GSov}$

$C_{GD} \approx C_{GDov}$

Asymmetric channel shape due to the pinch-off

C_{OX} is not distributed evenly between drain and source

Exercise 2 – Task 1.3

- 1.3 Simulate the transistor configuration shown in Figure 1 in order to extract C_{GS} and C_{GD} in the linear and saturation region. Compare the extracted values with the results of your calculation in task 1.1.

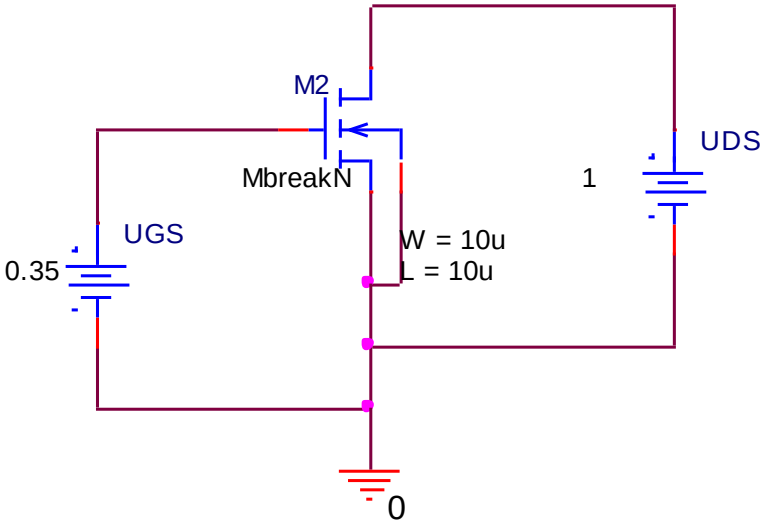
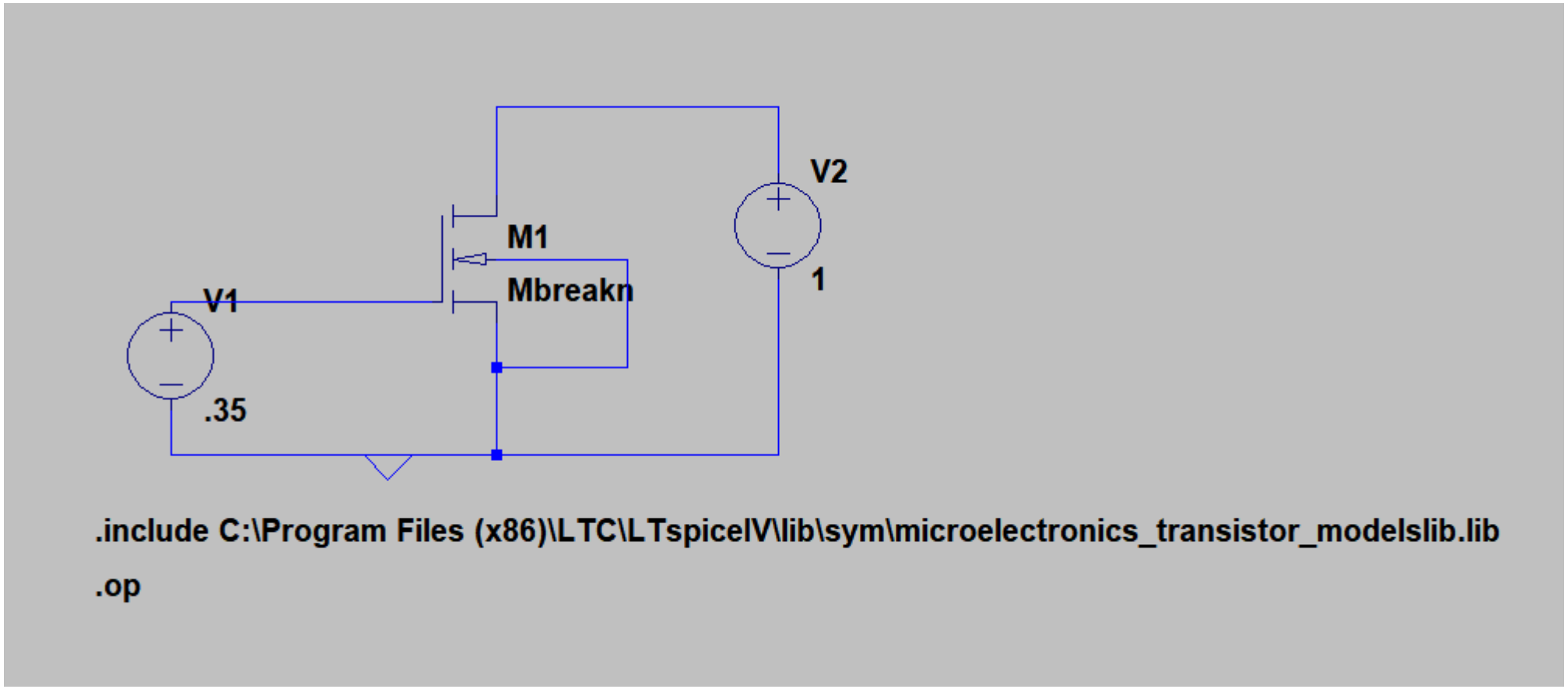


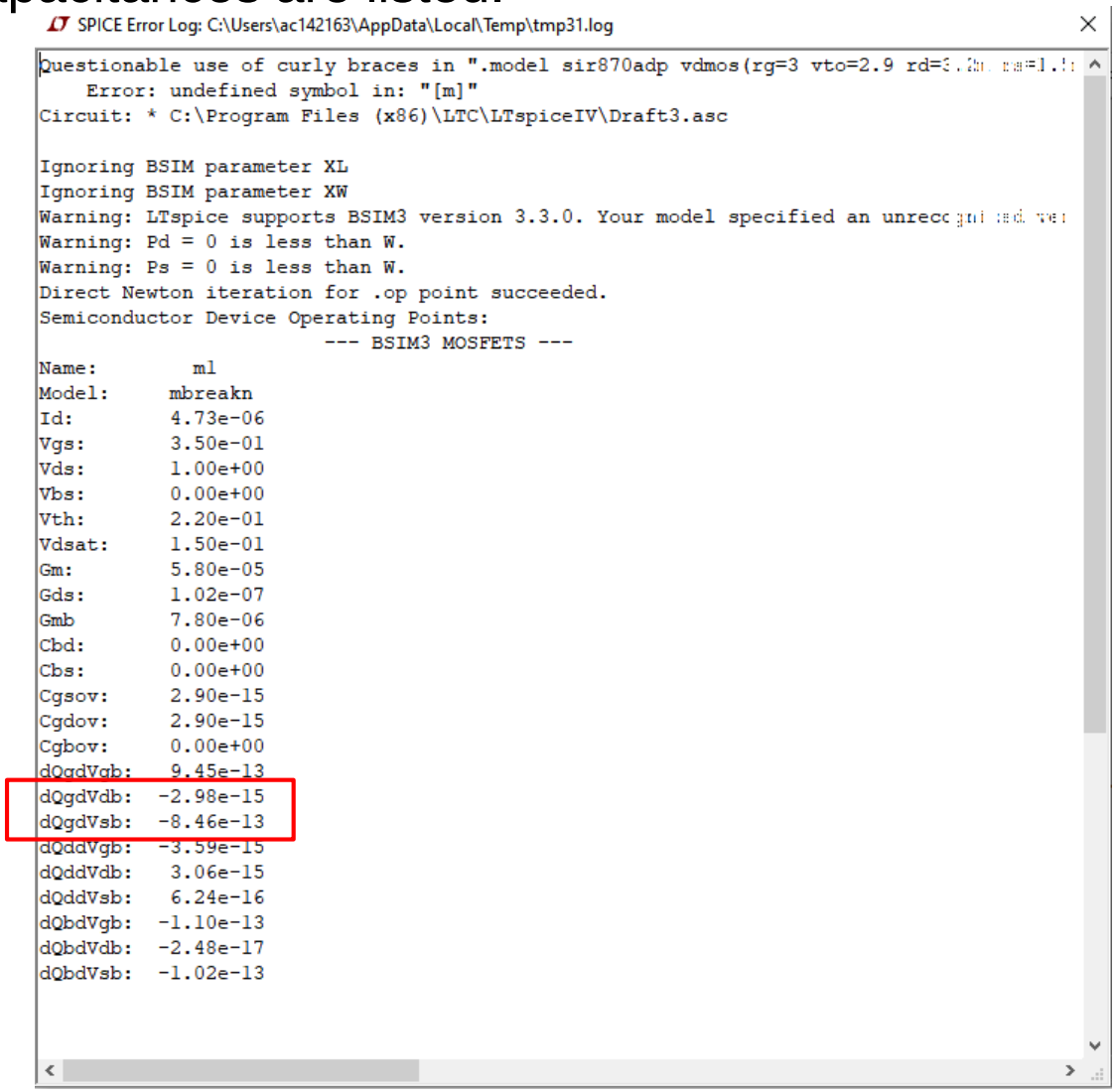
Figure 1: Biasing of an NMOS-transistor with variable U_{GS} and U_{DS}

- **How to extract small-signal parameters and capacitances in LTspice?**
- Set-up a DC operating point simulation
- *LTspice* → *Simulate* → *Edit Simulation Command* → select tab *'DC op pnt'*
- and perform the simulation.
- Once the simulation finished, open the SPICE Error Log file
- *LTspice* → *View* → *SPICE Error Log*

- and perform the simulation.



- where the small signal parameters and the capacitances are listed.
- The small signal parameters are labelled:
  - g_m → Gm
  - g_{DS} → Gds
  - g_{mb} → Gmb
- The capacitances are labelled:
  - C_{GS} → dQgdVsb
  - C_{GD} → dQgdVdb



Exercise 2 – Task 1.3

- 1.3 Simulate the transistor configuration shown in Figure 1 in order to extract C_{GS} and C_{GD} in the linear and saturation region. Compare the extracted values with the results of your calculation in task 1.1.

	linear region		saturation region	
	simulation	calculation (task 1.1)	simulation	calculation (task 1.1)
C_{GS}	0.723 pF	0.6 pF	0.846 pF	0.72 pF
C_{GD}	0.462 pF	0.6 pF	0.298 fF	0 pF

Simulaton settings:

- linear region $V_{GS} = 0.35\ \text{V}, V_{DS} = 0.02\ \text{V}$
- saturation region $V_{GS} = 0.35\ \text{V}, V_{DS} = 1\ \text{V}$

The results of the simulation and the calculation match quite well, i.e., the order of magnitude is correctly predicted by the calculation.

Exercise 2 – Task 1.4

- 1.4 In Figure 2, an RC low-pass filter is shown that can be used to extract the gate capacitance C_{OX}, if a calculation is not feasible due to an unknown C'_{ox}. Extract the corner frequency ω_c and calculate C_{OX} according to $\omega_c = \frac{1}{R\ C_{ox}}$. Compare the extracted C_{OX} with the results of your calculation in task 1.1.

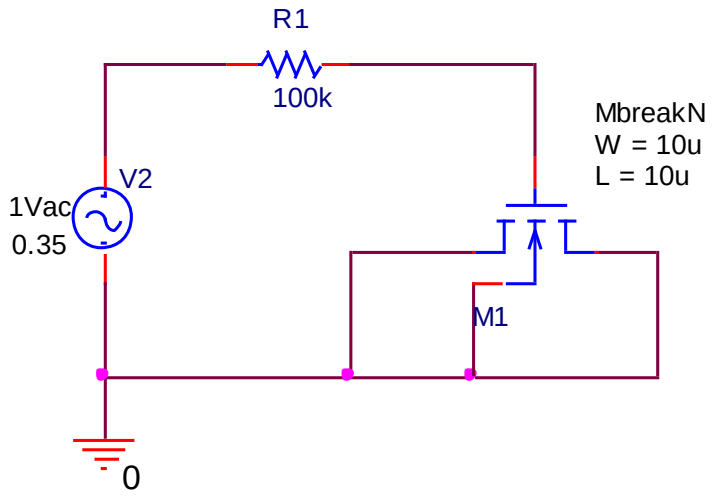


Figure 2: Low-pass filter for determination of C_{ox}

Exercise 2 – Task 1.4

- 1.4 In Figure 2, an RC low-pass filter is shown that can be used to extract the gate capacitance C_{OX}, if a calculation is not feasible due to an unknown C'_{ox}. Extract the corner frequency ω_c and calculate C_{OX} according to $\omega_c = \frac{1}{R\,C_{ox}}$. Compare the extracted C_{OX} with the results of your calculation in task 1.1.

$$f_c = 1.26\ \text{MHz} \rightarrow \omega_c = 2\pi f_c = 7.9168\ \text{Mrad/s}$$

$$C_{ox} = \frac{1}{R\omega_c} = \frac{1}{100\cdot 10^3 \cdot 2\pi \cdot 1.26\times 10^6} = 1.263\ \text{pF}$$

	simulation	calculation (task 1.1)
C_{Ox}	1.263 pF	1.2 pF

The results of the simulation and the calculation match quite well.

Exercise 2 – Task 2.1

- 2.1 Sketch the small-signal equivalent circuit of the circuit shown in Figure 3, including all parasitic capacitances. The transistors are operating in the saturation region.

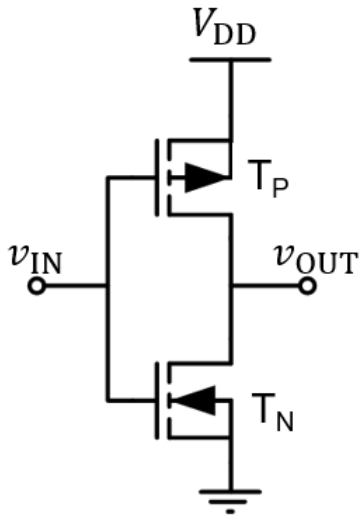
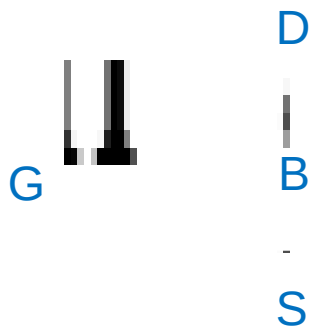


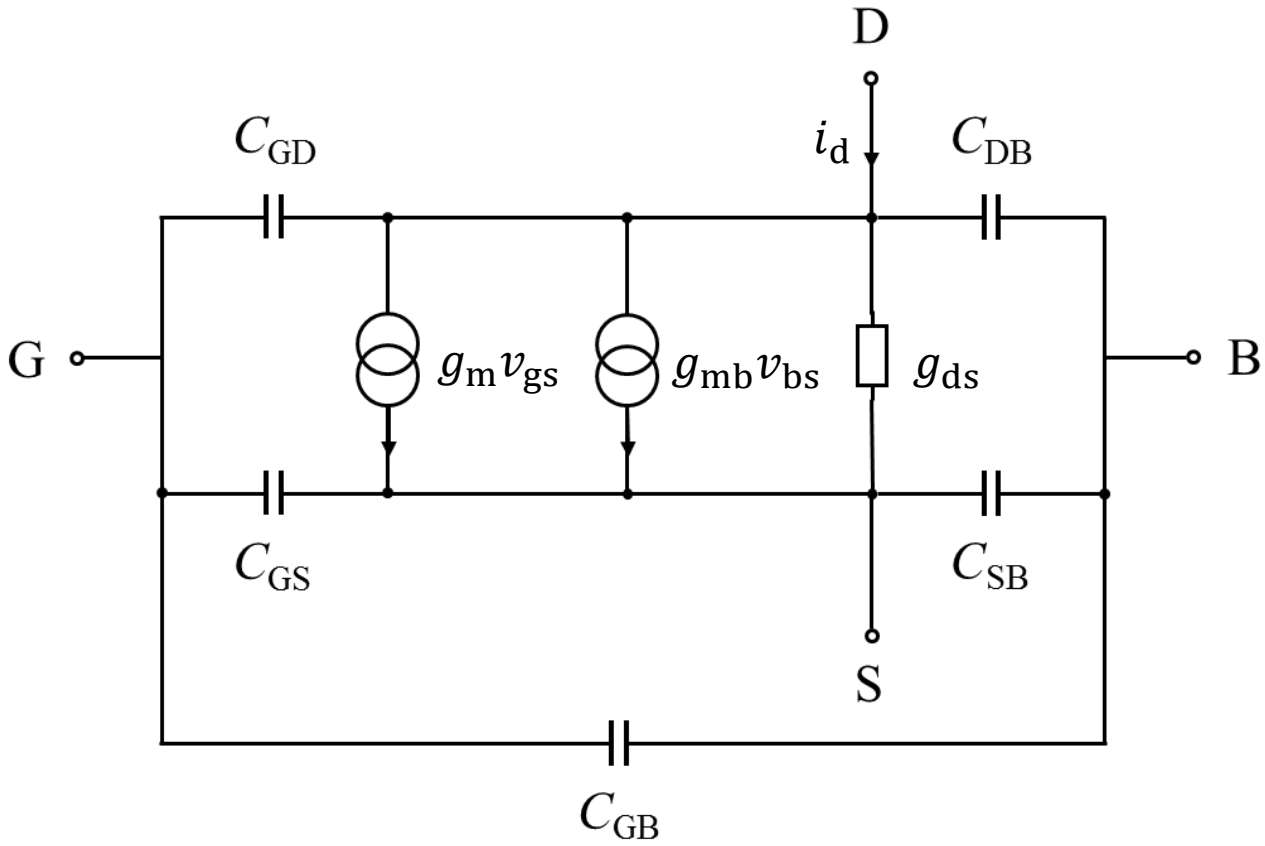
Figure 3: Single-stage amplifier

Exercise 2 – Task 2.1

- 2.1 Sketch the small-signal equivalent circuit of the circuit shown in Figure 3, including all parasitic capacitances. The transistors are operating in the saturation region.

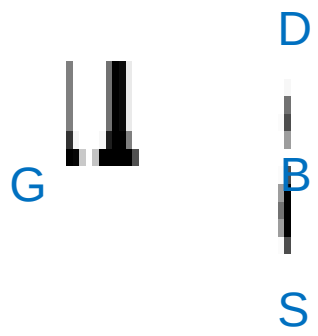
According to the lecture, the small-signal model of a MOS-transistor results in

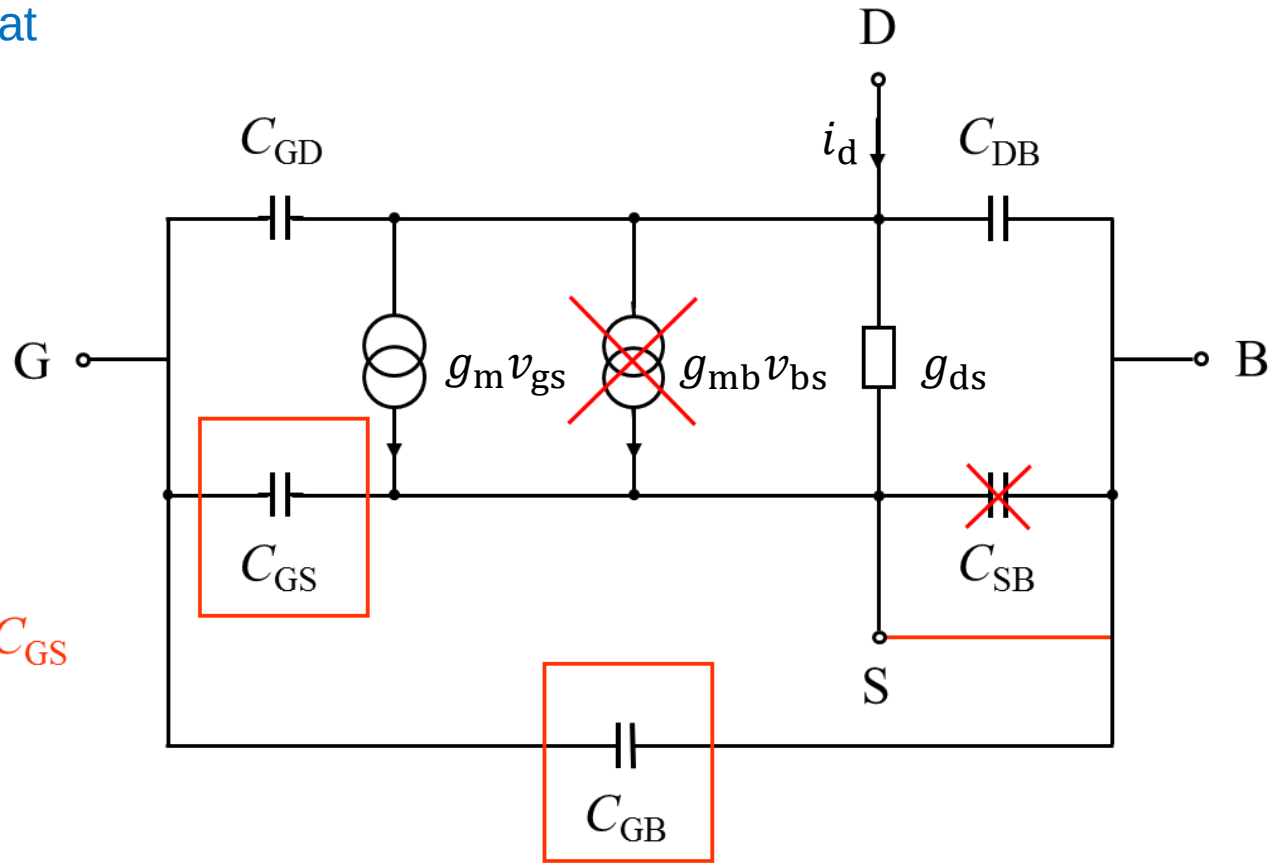




Exercise 2 – Task 2.1

For $V_{SB} = 0$ V, it holds that

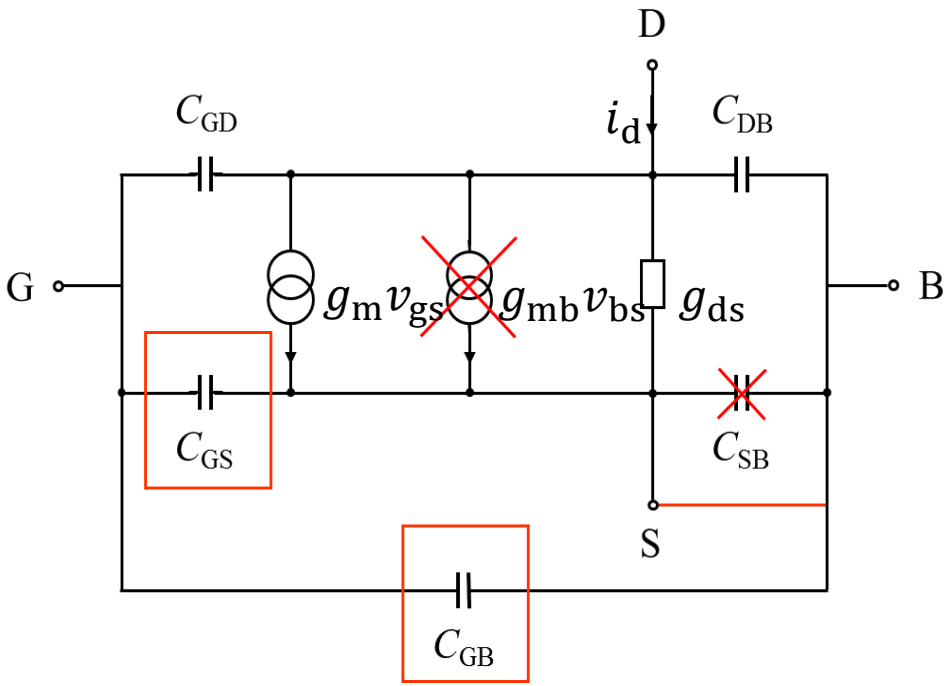


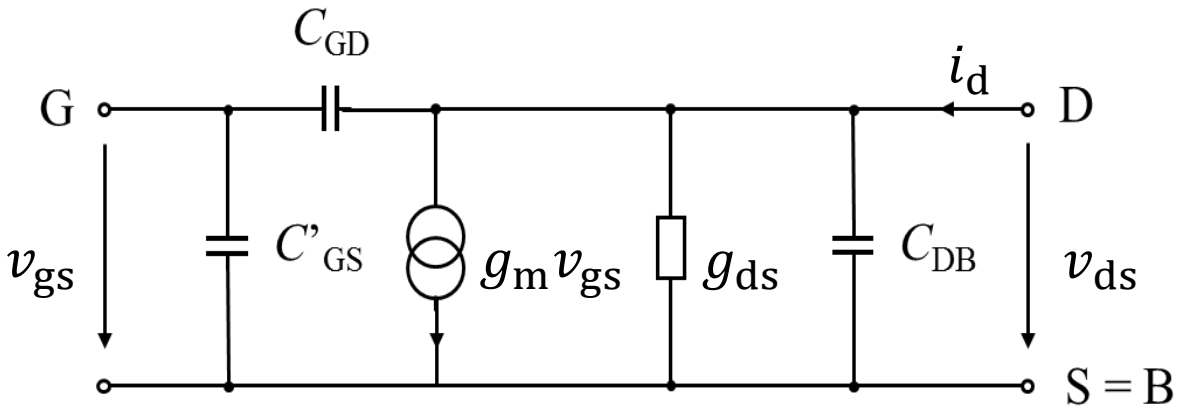


$$C'_{GS} = C_{GS} + C_{GB} \approx C_{GS}$$

Exercise 2 – Task 2.1

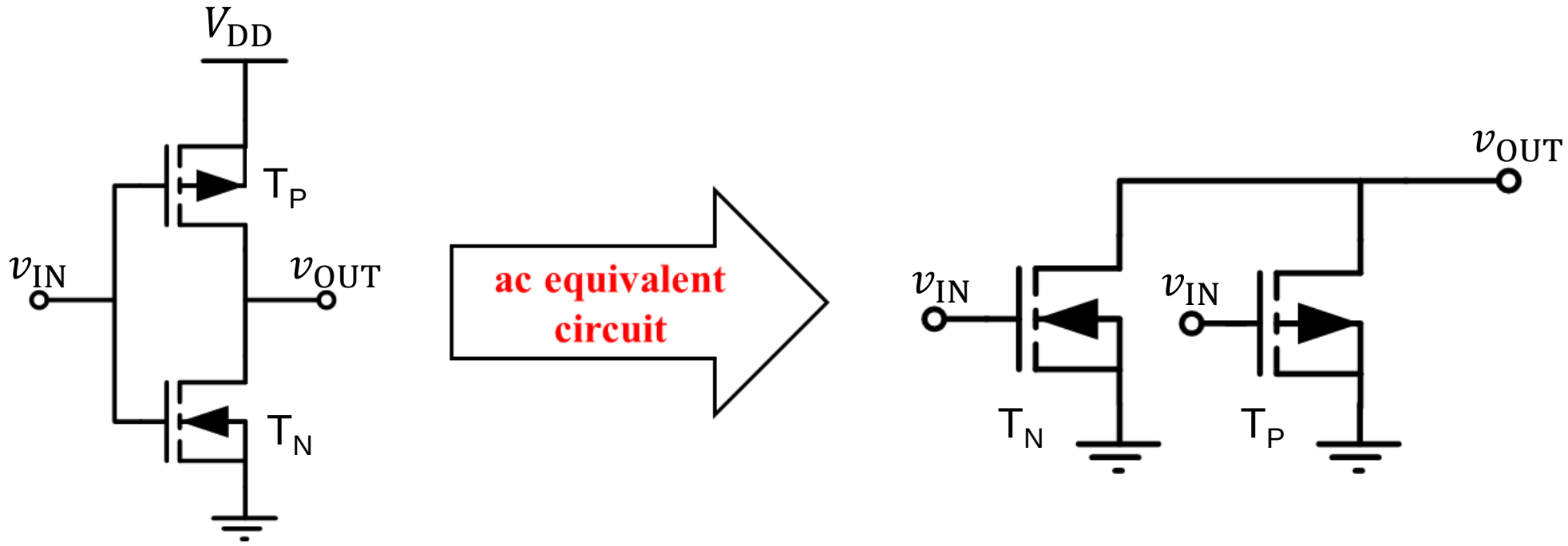
Whereby the small-signal model simplifies to





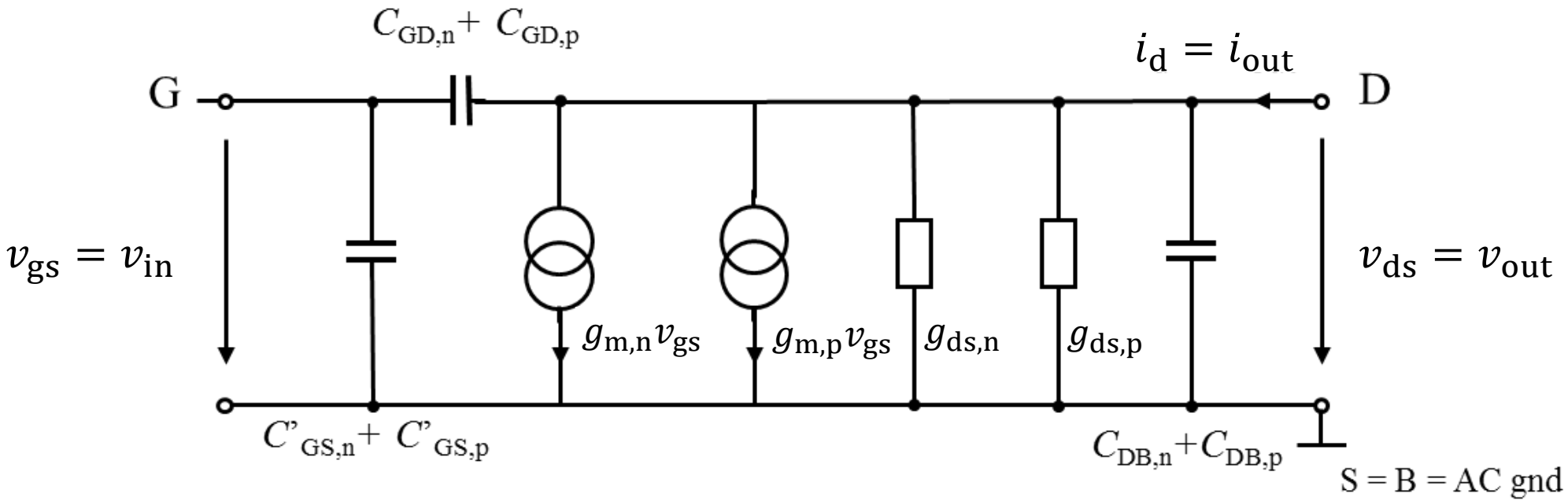
Exercise 2 – Task 2.1

For the derivation of the small-signal equivalent circuit, the single-stage amplifier shown in Figure 3 can be redrawn as



Exercise 2 – Task 2.1

Since any DC voltage source, i.e., the supply voltage V_{DD}, will be shorted. Thus, it can be seen that the PMOS-transistor is connected parallel to the NMOS-transistor. Accordingly, the small-signal equivalent circuit results in

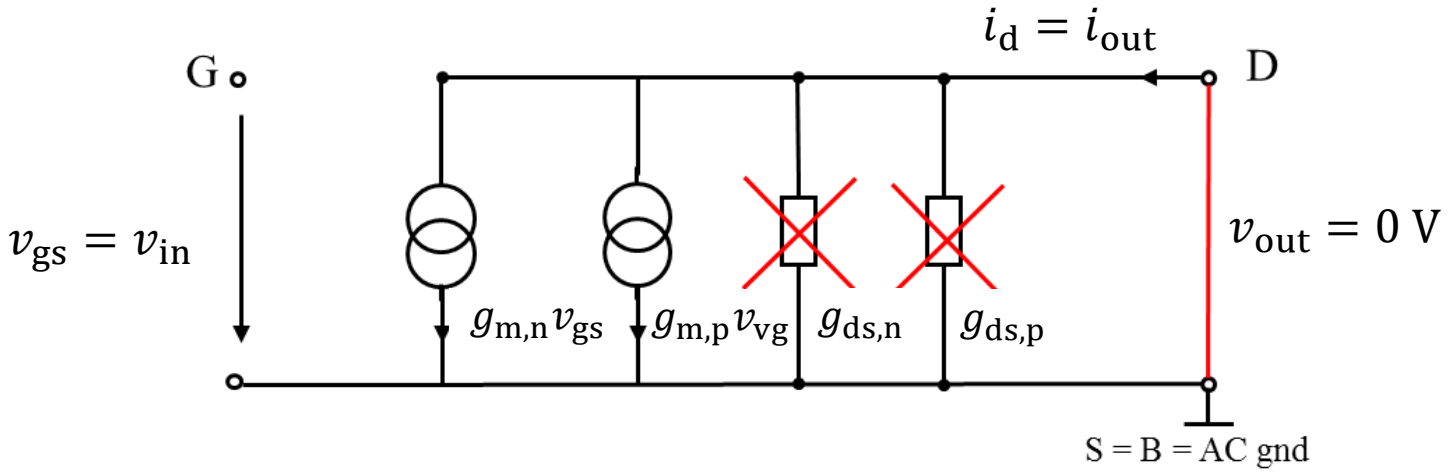


Exercise 2 – Task 2.2

- 2.2 Derive the small-signal transfer function $v_{\mathrm{OUT}}/v_{\mathrm{IN}}$ and thus the small-signal voltage gain! The parasitic capacitances are to be neglected.

Determination G_{m} (Y_{21})

In order to determine the transconductance Y_{21}, a short must be applied to the output of the small-signal model for v_{out} = 0 V and the ratio $i_{\mathrm{out}}/v_{\mathrm{out}}$ be determined.

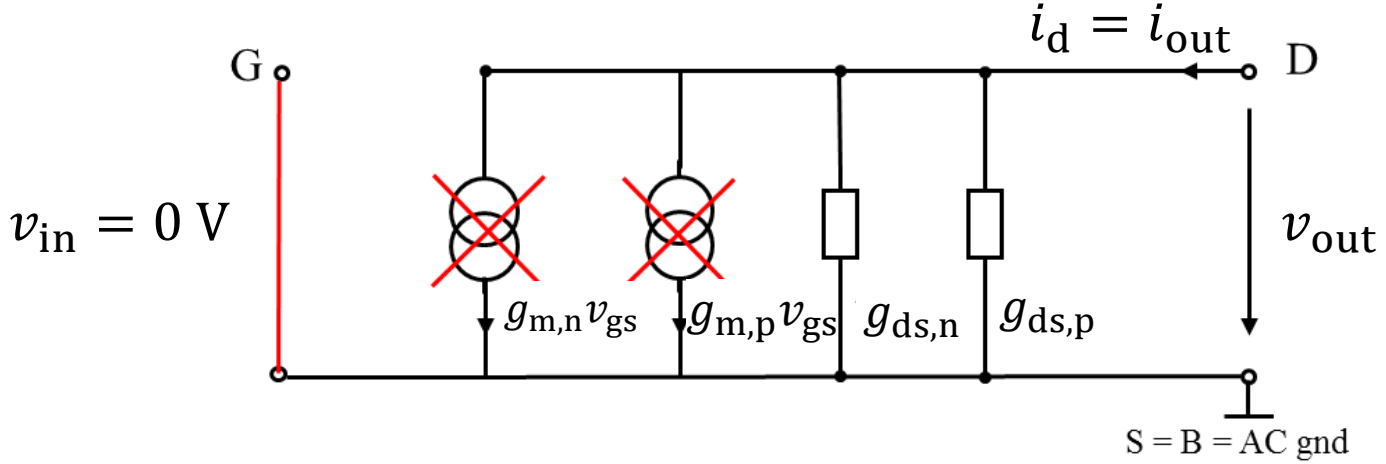


$$i_{\mathrm{out}} = g_{\mathrm{m,n}}v_{\mathrm{in}} + g_{\mathrm{m,p}}v_{\mathrm{in}} \rightarrow Y_{21} = G_{\mathrm{m}} = \frac{i_{\mathrm{out}}}{v_{\mathrm{in}}} = g_{\mathrm{m,n}} + g_{\mathrm{m,p}}$$

Exercise 2 – Task 2.2

Determination

In order to determine the transconductance Y_{22}, a short must be applied to the input of the small-signal model for v_{in} = 0 and the ratio i_{out}/v_{out} be determined.



$$i_{out} = \left(\frac{1}{r_{ds,n} \| r_{ds,p}}\right) v_{out} \rightarrow Y_{22} = g_{out} = \frac{i_{out}}{v_{out}} = \left(\frac{1}{r_{ds,n} \| r_{ds,p}}\right)$$

$$A = -\frac{Y_{21}}{Y_{22}} = -G_m r_{out} = -(g_{m,n} + g_{m,p})(r_{ds,n} \parallel r_{ds,p})$$

Exercise 2 – Task 2.3

- 2.3 Derive the output voltage swing, i.e., the voltage range in which both transistors are operating in the saturation region.

Since both transistors must operate in saturation, it holds that

$v_{\mathrm{DSn}} \geq v_{\mathrm{GSn}} - V_{\mathrm{thn}}$ and $|v_{\mathrm{DSp}}| \geq |v_{\mathrm{GSp}}| - |V_{\mathrm{thp}}|$

With $v_{\mathrm{GSn}} = v_{\mathrm{IN}}$ and $v_{\mathrm{DSn}} = v_{\mathrm{OUT}}$, it follows that

$$v_{\mathrm{OUT}} \geq v_{\mathrm{IN}} - V_{\mathrm{thn}}$$

Likewise: With $|v_{\mathrm{GSp}}| = V_{\mathrm{DD}} - v_{\mathrm{IN}}$ and $|v_{\mathrm{DSp}}| = V_{\mathrm{DD}} - v_{\mathrm{OUT}}$, it follows that

$$v_{\mathrm{OUT}} \leq v_{\mathrm{IN}} + |V_{\mathrm{thp}}|$$

The output voltage swing thus results in

$$v_{\mathrm{IN}} - V_{\mathrm{thn}} \leq v_{\mathrm{OUT}} \leq v_{\mathrm{IN}} + |V_{\mathrm{thp}}|$$

Exercise 2 – Task 2.4.1

- 2.4.1 Calculate the W/L-ratios of the NMOS- and PMOS-transistor for which the drain current results in 88.2 µA. Channel-length modulation is to be neglected.

The following parameters are given

$$V_{\mathrm{OUT}} = V_{\mathrm{IN}} = \frac{V_{\mathrm{DD}}}{2},\ V_{\mathrm{DD}} = 1.8\ \mathrm{V},\ \beta_{0\mathrm{n}} = 360\ \frac{\mu\mathrm{A}}{\mathrm{V}^2},\ V_{\mathrm{thn}} = 0.2\ \mathrm{V},\ \lambda_{\mathrm{n}} = 0.09\ \mathrm{V}^{-1}$$

$$\beta_{0\mathrm{p}} = 60\ \frac{\mu\mathrm{A}}{\mathrm{V}^2},\ V_{\mathrm{thp}} = 0.2\ \mathrm{V},\ \lambda_{\mathrm{p}} = 0.02\ \mathrm{V}^{-1}$$

$$I_{\mathrm{D}} = \frac{\beta_{0\mathrm{n}}}{2}\left(\frac{W}{L}\right)_{\mathrm{n}}\left(V_{\mathrm{GSn}} - V_{\mathrm{thn}}\right)^2 = \frac{\beta_{0\mathrm{p}}}{2}\left(\frac{W}{L}\right)_{\mathrm{p}}\left(|V_{\mathrm{GSp}}| - |V_{\mathrm{thp}}|\right)^2 \rightarrow \left(\frac{W}{L}\right)_{\mathrm{n}} = 1 \quad \& \quad \left(\frac{W}{L}\right)_{\mathrm{p}} = 6$$

Exercise 2 – Task 2.4.2

- 2.4.2 Calculate the small-signal parameters of the transistors, i.e., g_{m} and r_{DS}, using the W/L-ratios determined in task 2.4.1.

Since the transistors are operating in the saturation region, the small-signal parameters result in

$$g_{\mathrm{m,n}} = \left(\frac{W}{L}\right)_{\mathrm{n}} \beta_{0\mathrm{n}} \left(V_{\mathrm{GSn}} - V_{\mathrm{thn}}\right) = \sqrt{2\beta_{0\mathrm{n}} \left(\frac{W}{L}\right)_{\mathrm{n}} I_{\mathrm{D}}} = 252\ \mu S, \qquad r_{\mathrm{ds,n}} = \frac{1}{\lambda_{\mathrm{n}} I_{\mathrm{D}}} = 126\ \mathrm{k}\Omega$$

$$g_{\mathrm{m,p}} = \left(\frac{W}{L}\right)_{\mathrm{p}} \beta_{0\mathrm{p}} \left(|V_{\mathrm{GSp}}| - |V_{\mathrm{thp}}|\right) = \sqrt{2\beta_{0\mathrm{p}} \left(\frac{W}{L}\right)_{\mathrm{p}} |I_{\mathrm{D}}|} = 252\ \mu S, \qquad r_{\mathrm{ds,p}} = \frac{1}{\lambda_{\mathrm{p}} I_{\mathrm{D}}} = 567\ \mathrm{k}\Omega$$

Exercise 2 – Task 2.4.3

- 2.4.3 Calculate the small-signal voltage gain of the circuit in V/V and dB using the results of tasks 2.2 and 2.4.2.

With the small-signal voltage gain calculated in task 2.2, i.e.,

$$A_{\mathrm{inv}} = -\left(g_{\mathrm{m,n}} + g_{\mathrm{m,p}}\right)\left(\mathrm{r_{DS,n}} \parallel \mathrm{g_{DS,p}}\right)$$

and the small signal parameters of the transistor calculated in task 2.4.2

	NMOS	PMOS
g_{m}	$g_{\mathrm{m,n}} = 252\,\frac{\mu\mathrm{A}}{\mathrm{V}}$	$g_{\mathrm{m,p}} = 252\,\frac{\mu\mathrm{A}}{\mathrm{V}}$
r_{ds}	$\mathrm{r_{ds,n}} = 126\ \mathrm{k\Omega}$	$r_{\mathrm{ds,p}} = 567\ \mathrm{k\Omega}$

$$\rightarrow A_{\mathrm{inv}} = -(252 + 252)\,\frac{\mu\mathrm{A}}{\mathrm{V}}\ (126\ \mathrm{k\Omega} \parallel 567\ \mathrm{k\Omega}) = -51.96\,\frac{\mathrm{V}}{\mathrm{V}} \approx -52$$

$$20\log|A_{\mathrm{inv}}| = 20\log(52) = 34.32\ \mathrm{dB}$$

Exercise 2 – Task 2.4.4

- 2.4.2 Calculate the output voltage swing of the circuit using the results of task 2.3.

With the result of task 2.3, i.e.,

$$v_{\mathrm{IN}} - V_{\mathrm{thn}} \leq v_{\mathrm{OUT}} \leq v_{\mathrm{IN}} + |V_{\mathrm{thp}}|$$

the output voltage swing results in

$$(0.9\ \mathrm{V} - 0.2\ \mathrm{V}) \leq v_{\mathrm{OUT}} \leq (0.9\ \mathrm{V} + 0.2\ \mathrm{V}) \Rightarrow 0.7\ \mathrm{V} \leq v_{\mathrm{OUT}} \leq 1.1\ \mathrm{V}$$